5.6 Exploring Solutions to Systems of Equations

Lesson Introduction

Benchmark	MA.912.AR.9.1 Given a mathematical or real-world context, write and solve a system of two-variable linear equations algebraically or graphically.	Learning Target	• I can relate the solution of a system of equations to its graph and equations.

	Building On	Working Toward
Benchmark Coherence	MA.8.AR.4.1 Given a system of two linear equations and a specified set of possible solutions, determine which ordered pairs satisfy the system of linear equations. MA.8.AR.4.2 Given a system of two linear equations represented graphically on the same coordinate plane, determine whether there is one solution, no solution, or infinitely many solutions. MA.8.AR.4.3 Given a mathematical or real-world context, solve systems of two linear equations by graphing.	MA.912.AR.9.2 Given a mathematical or real-world context, solve a system consisting of a two-variable, linear equation, and a non-linear equation algebraically or graphically. MA.912.AR.9.3 Given a mathematical or real-world context, solve a system consisting of two-variable linear or non-linear equations algebraically or graphically.

Essential Question	• How is the solution of a system of equations related to its graph and equations?
Lesson Narrative	In this lesson, students are introduced to the concept of a system of equations. Students will bring together what they've learned about two variable equations and connect that to systems of two variable equations. They will spend this lesson laying the conceptual foundation for what a solution system is, and how it is represented graphically and verified through the system of equations.

Component Summary	5.6.1 Warm-Up (~5 min)	Mindsets in math activity *(SE p. 254)*	5.6.4 Bring It Together (10 min)	Systems of equations summary activity *(SE p. 259)*
	5.6.2 Exploration (15-20 min)	Solutions to systems Exploration activity *(SE p. 254)*	5.6.5 Wrap-Up (~5 min)	Self-reflection table activity *(SE p. 260)*
	5.6.3 Exploration (15-20 min)	Equivalent systems of equations *(SE p. 257)*		

	Glossary:	Materials	Additional Preparation
Resources	Key Terms: • System of Equations • Equivalent Systems Additional Terms: N/A	• (Optional) 5.6.1 Warm-Up contains a video to accompany the question prompts. • (Optional) Graphing utility	• (Optional) Vocabulary strategy such as a Frayer model

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may not recognize systems as equivalent. Systems that “look” different may be mistaken as different systems. This misconception highlights a lack of conceptual understanding in multiple representations or equivalence.</td><td>5.6.2 Exploration is an activity that is intended to review what students learned in the previous course before extending the exploration to equivalent systems.
This lesson is comprised primarily of explorations. Consider adding some time at the end of each exploration to review student findings through a class discussion.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
5.6.2 and 5.6.3 Explorations provide collaborative exploration opportunities with this instructional routine built into the activities. Students will have time to work through the exploration pieces individually and then collaborate with their partner.</td><td>MA.K12.MTR.1.1: Participate in Effortful Learning
This lesson consists primarily of exploration activities that require the students to explore and discover without instructor prompting. Through these activities, students will ask questions that will help with solving the task and help and support each other, while attempting to learn a new concept.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider the use of a digital graphing tool, such as Desmos, to show individual equations being graphed independently to highlight the solution of a system when the two equations exist on the same coordinate plane. This can be especially helpful when reviewing the information about equivalent systems.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

5.6.1 Warm-Up: Mindsets in Math

Component Overview: *Students will view a video about growth mindset in math and respond to three questions.*

5.6.1 Warm-Up: Mindsets in Math

1. Some people believe the myth that math intelligence is something people are only born with. This is called having a "fixed mindset." Others recognize that math intelligence is something anyone can develop, which is called having a "growth mindset."

   a. Describe something you have learned in math this year that you didn't initially understand, but you do now.
   Answers will vary.

   b. What helped you learn it?
   Answers will vary.

   c. How does your mindset regarding math intelligence impact the way you learn and face challenges in math class?
   Answers will vary.

Use growth mindset strategies throughout the lesson to continue to encourage growth mindset practices.

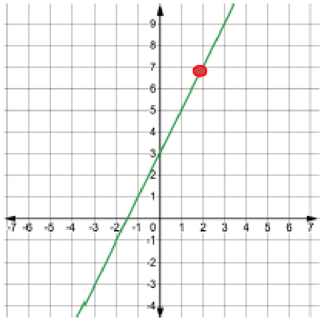
- *Positive narration and public recognition for students using growth mindset language*
- *Prompt students during the warm-up to look out for moments to practice growth mindset throughout the lesson; then have them reflect on their practices during the wrap-up.*

5.6.2 Exploration: Solutions to Systems of Equations

Component Overview: *Students will engage in an exploration of the solutions to systems of equations using five scaffolded, multipart questions. For this activity students will need a partner.*

5.6.2 Exploration: Solutions to Systems of Equations

1. The graph of the equation $y = 2x + 3$ is shown.



   a. Mark the point $(2, 7)$ on the graph. What do you notice?
   Answers will vary. The point is on the line.

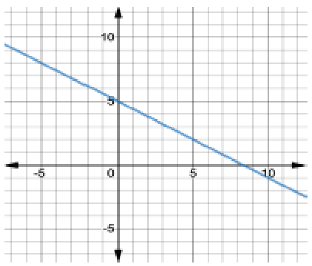
   b. Complete the statement.
   Since the point $(2, 7)$ lies on the line of $y = 2x + 3$, it is considered a(n)
   - o intercept
   - o solution
   - o vertex

   to the equation.

   c. Verify algebraically that point $(2, 7)$ is a solution to the equation $y = 2x + 3$.
   $7 = 2(2) + 3$
   $7 = 4 + 3$
   $7 = 7$

This activity is an exploration that is intended to review what students learned in the previous course before extending the exploration to equivalent systems. Consider exploring this activity as a student first to better inform how to facilitate sensemaking and connections during this student-centered activity.

2. The graph of the equation $y = -0.6x + 5$ is shown.



Why is it important to verify that the solution is correct?

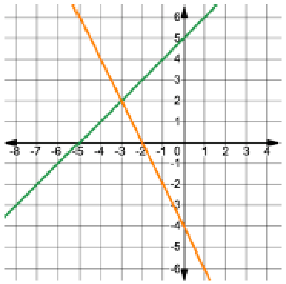
5.6.2 Exploration: Solutions to Systems of Equations (continued)

Work with your partner to find two different solutions to the equation $y = -0.6x + 5$. Then, verify algebraically that each is a solution.

Solution 1: (0 , 5)	**Solution 2: (5 , 2)**
Algebraic Verification $5 = -0.6(0) + 5$ $5 = 0 + 5$ $5 = 5$	Algebraic Verification $2 = -0.6(5) + 5$ $2 = -3 + 5$ $2 = 2$

For question 2, students will need a partner to complete the task for the remainder of the activity.

3. The graphs of two linear equations are shown.



   a. Discuss with your partner whether or not there is a common solution for both linear equations shown on the graph.

Why do you think it's possible to have more than one solution for this equation?

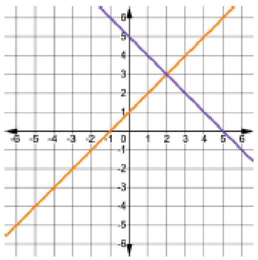
   b. Complete the statement.

   The point (−3, 2) lies on both lines, so it is considered a solution of both linear equations.

Monitor for student responses that highlight the point resting on both lines and language like, "both lines sharing the point $(-3,2)$." Highlight those responses whole-group using talk moves.

A ***system of equations*** is a set of two or more equations. Each equation contains two or more variables. The solution is all the values for the variables that make all equations in the system true.

4. The graph of the system of equations, $\begin{cases} y = x + 1 \\ y = -x + 5 \end{cases}$, is shown.



   a. Work with your partner to determine the solution to the system of equations from the graph.

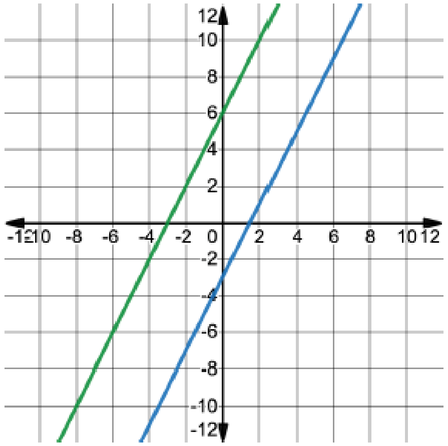
5.6.2 Exploration: Solutions to Systems of Equations (continued)

(2,3)

b. Verify algebraically that your solution satisfies both equations in the system.

$y = x + 1$	$y = -x + 5$
$3 = 2 + 1$ $3 = 3$	$3 = -2 + 5$ $3 = 3$

5. The graph of a system of equations is shown.



Work with your partner to describe the solution of the system.
There is no solution.

This will be the students' first time learning about the concept of a system of equations. After the activity, consider having students use a vocabulary strategy - such as a Frayer model - to highlight the term, what it is, and what it isn't, so that students have a working knowledge of the term as they progress through the remainder of this unit.

Why do you think this system only has one solution and not multiple like a single equation?

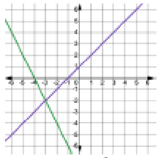
If it was one line instead of two, would it still be no solution? Why or why not?

5.6.3 Exploration: Equivalent Systems of Equations

Component Overview: *Students will explore the concept of equivalent systems of equations through one multipart scaffolded question. For this activity, students will need a partner.*

5.6.3 Exploration: Equivalent Systems of Equations

1. Chloe, Armin, and Olivia are all solving the system of equations, $\begin{cases} y = -2x - 8 \\ y = x + 1 \end{cases}$. The graph of the system is shown.



   a. What is the solution to this system?

   $(-3, -2)$

Each student rewrote the original system of equations in different ways. Their new systems and how they generated them are shown in the table.

Student	Student's Work	New System
Chloe	$y = -2x - 8$ $\underline{+2x \quad +2x}$ $2x + y = -8$ $y = x + 1$ $\underline{-x \quad -x}$ $-x + y = 1$	$\begin{cases} 2x + y = -8 \\ -x + y = 1 \end{cases}$
Armin	$y = -2x - 8$ $\underline{+(y = x + 1)}$ $2y = -x - 7$ $\underline{+x \quad +x}$ $x + 2y = -7$ $y = x + 1$ $\underline{-x \quad -x}$ $-x + y = 1$	$\begin{cases} x + 2y = -7 \\ -x + y = 1 \end{cases}$
Olivia	$y = -2x - 8$ $\underline{+2x \quad +2x}$ $2x + y = -8$ $5(y = x + 1)$ $5y = 5x + 5$ $\underline{-5x \quad -5x}$ $-5x + 5y = 5$	$\begin{cases} 2x + y = -8 \\ -5x + 5y = 5 \end{cases}$

Consider beginning this activity by prompting students to use what they know. By highlighting the metacognition (thinking about thinking), you are encouraging practices of activating prior knowledge, while positioning students as experts of their own experience.

5.6.3 Exploration: Equivalent Systems of Equations (continued)

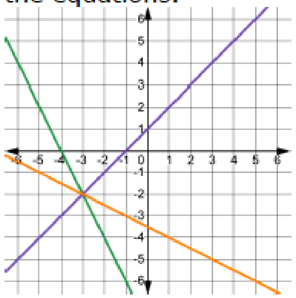
b. Chloe rearranged her equations. Explain what Armin did to generate his new system.
Armin added the two equations in the original system to create one equation. Then he rewrote the second equation into standard form.

c. Explain how Olivia generated her new system.
Olivia rewrote one equation into standard form. She multiplied the other equation by 5 and rewrote it into standard form.

d. With your partner, discuss whose new system you think will result in the same solution as the original system. Then, summarize your discussion.
Answers will vary.

The graphs of the original system of equations and the newly created systems are shown on the same coordinate grid. The graphs that represent each equation are shown in the same color as the equations.

$y = -2x - 8$
$2x + y = -8$



$y = x + 1$
$-x + y = 1$
$-5x + 5y = 5$

$x + 2y = -7$

The graphs of $y = x + 1$, $-x + y = 1$, and $-5x + 5y = 5$ are on the same line, shown in purple on the coordinate grid. Therefore, these equations are **equivalent**, not equal. To be equal, the equations would be exactly the same.

Consider the use of a digital graphing tool, such as Desmos, to show individual equations being graphed independently to highlight the solution of a system when the two equations exist on the same coordinate plane. This can be especially helpful here when reviewing the information about equivalent systems.

5.6.3 Exploration: Equivalent Systems of Equations (continued)

When two entities have the same result, they are **equivalent**. Systems of equations with the same solution are **equivalent systems**.

e. Complete the statement.

Chloe, Armin, and Olivia all generated systems of equations [○ equal / ○ equivalent] to the original system because all of the systems result in [○ the same / ○ different] solution(s).

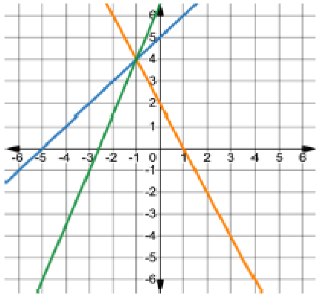
Consider adding some time at the end of 5.6.3 Exploration to review student findings through a class discussion.

5.6.4 Bring It Together: Systems of Equations

Component Overview: *Students will bring together what they have discovered in the prior two explorations to formalize the concept of a system of equations.*

5.6.4 Bring It Together: Systems of Equations

1. The graphs of the systems of equations $\begin{cases} 2x + y = 2 \\ x - y = -5 \end{cases}$ and $\begin{cases} 5x - 2y = -13 \\ x - y = -5 \end{cases}$ are shown on the same coordinate grid.



Complete the statements.

a. The solution of any system of equations is the
   - o point(s) of intersection.
   - o slope.
   - o y-intercept.

b. The solution of the system $\begin{cases} 2x + y = 2 \\ x - y = -5 \end{cases}$ is $(-1, 4)$.

c. The solution of the system $\begin{cases} 5x - 2y = -13 \\ x - y = -5 \end{cases}$ is $(-1, 4)$.

d. The systems of equations $\begin{cases} 2x + y = 2 \\ x - y = -5 \end{cases}$ and $\begin{cases} 5x - 2y = -13 \\ x - y = -5 \end{cases}$ are equivalent systems because they have the same solution.

2. Verify algebraically that the solution satisfies all of the equations included in either system.

$2x + y = 2$	$x - y = -5$	$5x - 2y = -13$
$2(-1) + 4 = 2$ $-2 + 4 = 2$ $2 = 2$	$-1 - 4 = -5$ $-5 = -5$	$5(-1) - 2(4) = -13$ $-5 - 8 = -13$ $-13 = -13$

For this activity, have students work with their partner to complete the statements and answer the questions in this activity. At the conclusion of the activity, bring the class back together to engage in a collaborative discussion for each of the statements in this activity. This will help clear up any potential misconceptions and ensure students have a strong foundational understanding of this concept as the unit progresses.

Watch for student miscalculations with question 2. Without calculators, students tend to make errors when working with negative values.

5.6.5 Wrap-Up: Reflection

Component Overview: *Students will complete a self-reflection of what they learned in the exploration activities of this lesson.*

5.6.5 Wrap-Up: Reflection

1. Complete the table.
   Answers will vary.

The terms and information used in the Explorations that I knew were …	**Something I wondered during the Explorations was …**	**Something I learned during the Explorations was …**

Consider transforming this exercise into a collaborative exploration and application of growth mindset to connect to the warm-up. Have students find a partner and explain their responses. The partner must ask questions and take notes, and the teacher will then ask students questions about their partner's responses.